

FLM: A Fly-Wired Recurrent Language Model

An inspectable small-data experiment with matched lexical baselines

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Abstract

We implement a compact language model whose directed recurrent topology is selected from a fruit-fly connectome. A differentiable rate core with fast and slow state receives a learned text interface and predicts a lossless, training-only byte-pair vocabulary. The 600,003-parameter model is compared with a 595,408-parameter GRU and a 607,468-parameter decoder transformer on the official WikiText-2 raw partitions. All models share lexical dimensions, training windows and update budgets. The browser implementation exposes actual state, interventions and a separate online readout adapter. This report documents the implemented methods and preliminary validation evidence. The current matched result does not show an advantage over the conventional baselines. Final test-set scoring is gated on completion of all six registered runs. Anatomical identity, small recurrent state and browser accessibility are demonstrated engineering properties; improved data efficiency, biological learning and whole-brain language competence are not established.

1 Question and scope

Can an anatomical wiring constraint be a useful prior for language prediction under a small training budget? FLM makes this question concrete rather than treating a connectome as a pretrained intelligence. Training from scratch means that no learned language weights, pretrained embeddings, pretrained tokenizer or synthetic teacher corpus are imported. It does not mean absence of priors: graph selection, sign assumptions, pooling, nonlinearities and optimization all impose structure.

The intended contribution is an inspectable experimental system: reproducible graph and text acquisition, a compact recurrent architecture, matched baselines, exact browser parity, and visible separation between prediction, adaptation and body animation. A new name is not evidence of an algorithmic breakthrough. The current evidence supports a functioning next-token predictor and a testable architectural hypothesis, not a competitive general assistant.

2 Research foundations

MaleCNS provides anatomical connectivity and neuron identities [5, 3]. Simplified connectome-based dynamics have been used to investigate specific sensorimotor responses [10]; task optimization within anatomical constraints has precedent in models of fly visual computation [6]. Neither result licenses transfer of behavioral validity to a language-trained, central-brain subset with artificial interfaces.

Connectome-based reservoir computing is established prior work. Morra and colleagues studied a fly lateral-horn reservoir on a multifunctionality benchmark [8]. FLM differs from a fixed reservoir because it learns edge magnitudes and time constants as well as its input and readout. That difference must itself be evaluated; it is not inherently an improvement. Eligibility-trace learning provides another recurrent-learning direction [1], but FLM currently uses backpropagation through time, not a biological credit-assignment algorithm.

Transformers [11], gated recurrent models [2], SpikeGPT [13], RWKV [9] and Mamba [4] establish that attention-free or recurrent language prediction is not new. FLM’s narrow research question concerns this

graph prior and these memory filters under declared conditions. Sparse connectivity alone does not establish lower power consumption on conventional hardware.

3 Anatomy and graph construction

The pinned source snapshot contains 166,700 annotated neurons, 25,582,938 directed neuron-pair edges and 124,177,617 contacts. These are counts of the acquired runtime representation, not a claim that the compact model contains the complete biological circuit. File sizes, hashes, index ranges, orientation and contact sums are independently checked. The snapshot is identified in the repository’s source manifest.

The language core selects the 1,024 central-brain neurons with the largest incoming-plus-outgoing contact strength among the declared central intrinsic population, before inspecting language outcomes. Its induced graph retains 76,130 edges. Discarded boundary connections are absent; no unobserved external neurons are silently simulated. Type-ordered balanced pooling produces 128 pools. Five selected neurons lack finite soma coordinates and remain in inference but are omitted from positional rendering.

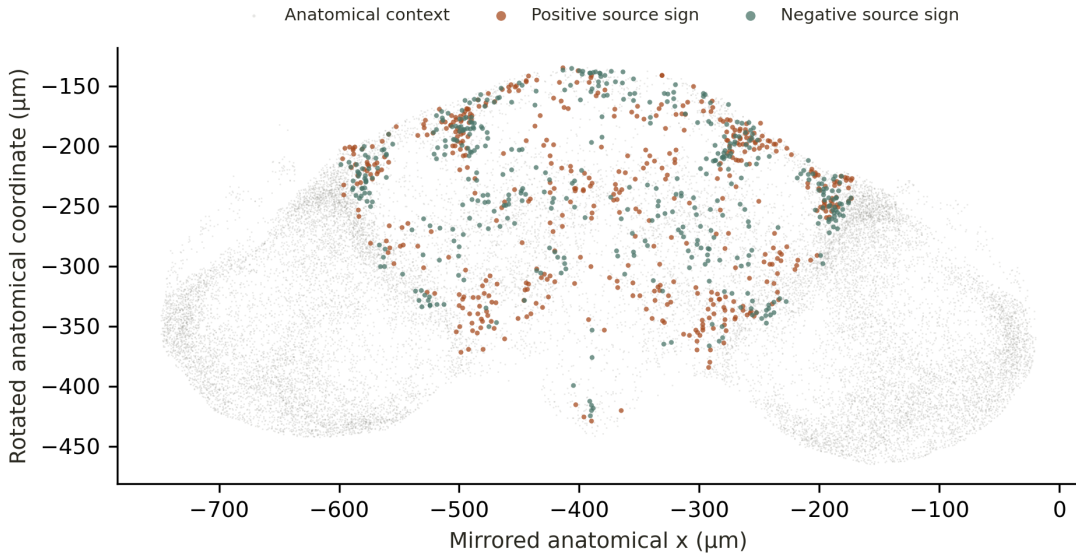


Figure 1: Anatomical positions of the selected neurons over downsampled brain context. Color denotes the assumed source sign, not activity. The displayed coordinates are rigidly rotated and mirrored for readability. The browser instead colors active neurons from actual inference state.

Signs are modeling assumptions associated with source neurotransmitter metadata. Contact counts are not measured functional weights. A conflict between an upstream model-card description and runtime metadata for modulatory transmitters is recorded; the declared runtime rule uses zero for those signs, and zero-sign outgoing edges are removed. The complete provenance and selection card are part of the reproducibility record.

4 Architecture

Let x_t be a token ID, $E \in \mathbb{R}^{4096 \times 96}$ a trainable embedding, $h_t, s_t \in \mathbb{R}^{1024}$ the fast and slow states, and P balanced mean pooling to 128 features. For retained edge $j \rightarrow i$, let c_{ij} be its contact count and σ_j its declared source sign. Define

$$v_{ij} = \sigma_j \log(1 + c_{ij}) \exp(\text{clip}(\theta_{ij}, -3, 3)), \quad (1)$$

$$W_{ij} = v_{ij} / \max\left(10^{-8}, \sum_k |v_{ik}|\right). \quad (2)$$

Missing edges remain zero. Positive multiplicative gains preserve sign, and incoming absolute normalization is repeated after learning the gains. The update is

$$u_t = AE[x_t] + b + gWh_{t-1}, \quad (3)$$

$$h_t = (1 - \alpha) \odot h_{t-1} + \alpha \odot \tanh(u_t), \quad (4)$$

$$s_t = (1 - \beta) \odot s_{t-1} + \beta \odot h_t, \quad (5)$$

$$f_t = \text{LN}([Ph_t; Ps_t]), \quad (6)$$

$$z_t = Rf_t + r, \quad \ell_t = Ez_t + q. \quad (7)$$

Layer normalization uses $\epsilon = 10^{-5}$. Learned bounds are $g = 0.05 + 2.95 \text{sigmoid}(\gamma)$, $\alpha_i = 0.05 + 0.90 \text{sigmoid}(a_i)$, and $\beta_i = 0.002 + 0.098 \text{sigmoid}(b_i)$. These are discrete token-update rates, not milliseconds or inferred membrane constants. The same embedding matrix receives gradients from input encoding and output scoring. There is no attention layer or direct input-to-logit bypass.

The state is causal and bounded by the leaky tanh dynamics from zero initialization. Bounded values do not guarantee long-context retention, a stable local Jacobian or a useful attractor. The slow filter adds a timescale, but its effective memory depends on recurrence, input and the learned readout. An isolated scalar filter with rate r has half-life $\log(0.5)/\log(1 - r)$; this is a descriptive filter property rather than a measured language-memory horizon.

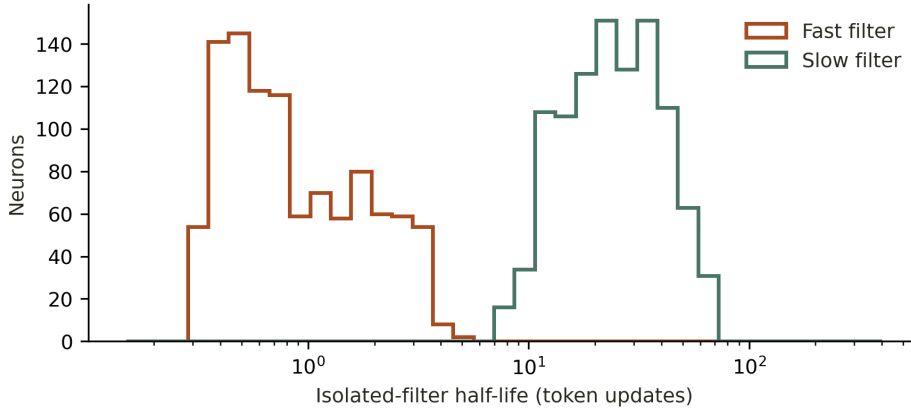


Figure 2: Learned isolated-filter half-lives at browser checkpoint 3000. Recurrent coupling and readout can change the effective task-level memory.

At this scale PyTorch uses a dense matrix containing the sparse pattern because it was faster on the available CPU. The browser uses destination-major compressed sparse rows and stores the tied embedding once, yielding a 2,713,236-byte inference-weight package. It computes input drives on demand instead of expanding all 4,096 tokens into 1,024-dimensional cached drives. Independent PyTorch fixtures verify logits, fast state, slow state and readout features.

5 Data and matched evaluation

WikiText was introduced to support language modeling with broader vocabulary and longer document structure [7]. We use the pinned raw variant, preserving official partitions of 600 training, 60 validation and 60 test articles. Training contains 2,051,910 whitespace-delimited words and 10,914,845 UTF-8 bytes. The raw variant still contains preprocessing artifacts such as punctuation markers; it is not an untouched Wikipedia dump.

A lossless byte-level BPE tokenizer is trained exclusively on the 600 training articles. It retains every byte through a complete byte alphabet and uses 4,094 text IDs plus explicit BOS=0 and EOS=1. No string in ordinary input is automatically treated as a boundary token. Training text averages 3.54 bytes per content token. Official split preservation, exact text reconstruction, hashes and limitations are detailed in the companion data note.

The transformer has pre-normalized residual blocks, feedforward width 216, GELU and rotary positions; causal attention retains a 96-position window per layer. Streaming uses native bounded key/value caches,

Model	Parameters	Core configuration
FLM	600,003	1,024 neurons, 128 pools, two states
GRU	595,408	One layer, hidden width 200
Transformer	607,468	Two layers, width 108, six heads

Table 1: All models share a 96-dimensional tied lexical interface and the same 4,096-token vocabulary. Each baseline is within 1.3% of FLM’s parameter count.

not repeated rescoring of the input prefix. The GRU uses its native recurrent hidden state. Their architecture-specific initialization remains part of the comparison and is recorded in source.

Each run uses 6,000 AdamW updates, batch 16 and sequences of 96 tokens. The first 16 positions warm the state and are excluded from the training loss. Sequences stay within articles. Training samples windows with replacement using a NumPy RNG independent of model initialization, so models with the same seed see identical inputs. Each run presents 9,216,000 input tokens. The learning rate warms up over 100 updates to 0.002 and decays by cosine to 0.0002; weight decay is 0.01 and gradient norm is clipped at 1.0. There is no dropout. Seeds 42 and 43 are registered for each main architecture.

Every 500 updates, validation uses deterministic per-article prefixes with a total quota of 32,768 target positions. State resets between articles and is carried within them. Boundary IDs are excluded from scored text. Selection minimizes validation bits per original UTF-8 byte:

$$\text{BPB} = \frac{-\sum_{t \in \mathcal{T}} \log p(x_t | x_{<t})}{\log 2 \sum_{t \in \mathcal{T}} \text{bytes}(x_t)}. \quad (8)$$

This measures the codelength of the canonical token sequence per byte. Shared-tokenizer perplexity is also reported; it must not be compared directly to published word-token WikiText perplexities. The same exact article bytes are required for paired final-test comparisons.

6 Preliminary evidence

The table reports seed 42 at the largest update count currently shared by all three published validation trajectories. It is a validation-prefix comparison, not a final held-out test result. The full six-run study is still being completed. Graph-specific superiority cannot be inferred without the declared rewired and disabled-mechanism controls.

Model	Parameters	Bits/byte	Token perplexity
FLM	600,003	1.9630	128.48
GRU	595,408	1.8913	107.58
Transformer	607,468	1.8870	106.45

Table 2: Matched validation at 3000 updates, seed 42. Lower is better. Values are generated directly from versioned evaluation artifacts.

The current FLM result trails the conventional baselines. Unedited same-prompt continuations show learned word and phrase structure, but also repetition, malformed names, preprocessing markers and unsupported statements. These outputs justify completion mode rather than an assistant claim. A few attractive continuations would not demonstrate factual knowledge or understanding.

The earlier AMI byte experiment provides a separate negative result: at 6,000 updates FLM obtained 1.9683 validation-prefix bits/byte, while a smoothed four-byte-context n-gram obtained 1.6091 on its corresponding byte-prefix protocol. AMI and WikiText scores describe different distributions and tokenization protocols and must not be placed on one quality leaderboard. The n-gram is not parameter-matched.

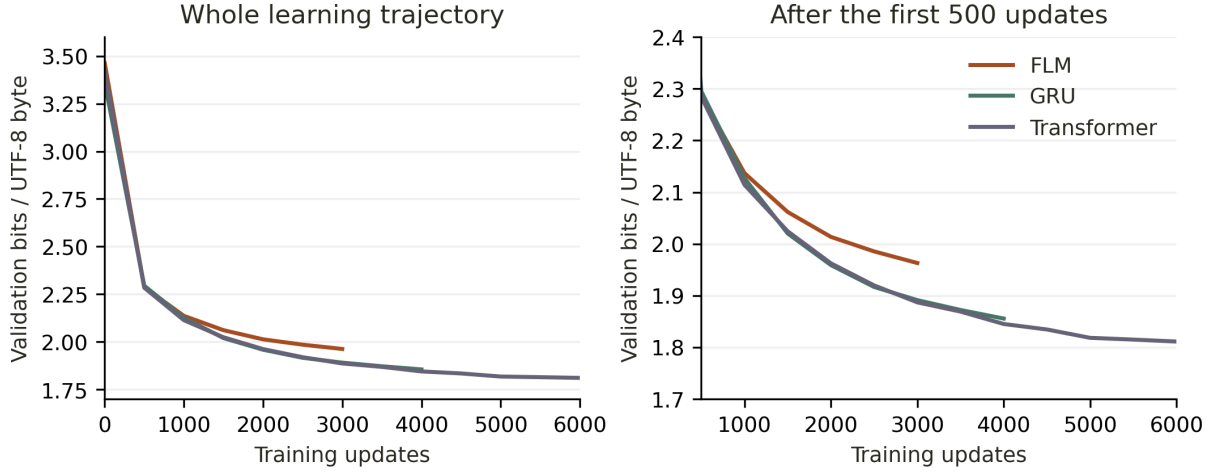


Figure 3: Actual validation trajectories. Unequal trajectory endpoints indicate unfinished runs, not equal-compute endpoints. Compare the same update count. The focused panel makes differences after early vocabulary acquisition visible.

7 Memory, adaptation and visible behavior

FLM stores 2,048 float32 state values, or 8,192 bytes per sequence. The GRU’s hidden state is smaller at 800 bytes. The declared transformer’s two-layer cache retains up to 95 past positions per layer and stores 164,160 float32 key/value bytes, excluding its scalar position counter. These calculations exclude weights, temporary activations, allocator overhead and adaptation. They are memory accounting, not speed or energy benchmarks.

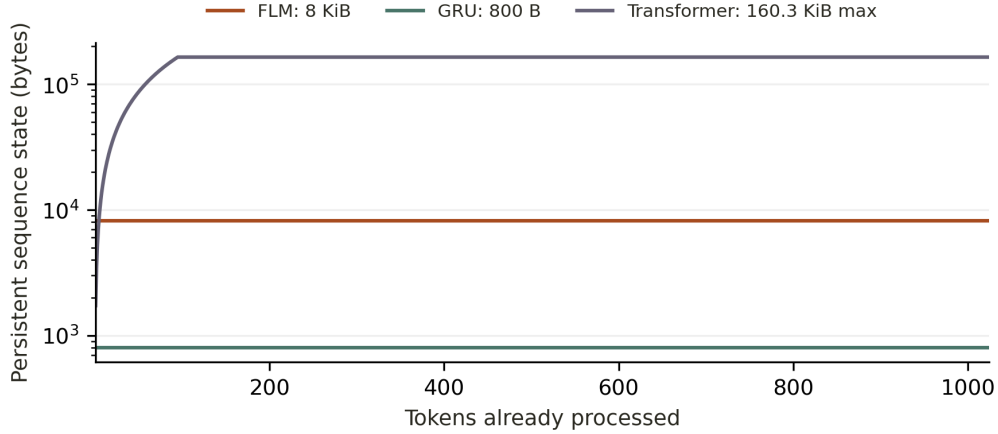


Figure 4: Persistent sequence-state storage derived from the implemented dimensions. The transformer’s cache is bounded; FLM is not the smallest-state comparator.

Browser learning updates a separate 4096×96 output correction and bias by the observed next-token cross-entropy gradient, with rate scaling by $1/\sqrt{96}$, small decay and bounded coefficients. It does not modify the bundled embedding, recurrent wiring or time constants. Consequently, a fixed input has the same recurrent activity before and after readout adaptation; free generation can follow a different trajectory because its selected input tokens change. Separate probe text is scored before and after updates. Lower training loss alone may be memorization.

The 3D brain displays actual fast or slow state for modeled neurons. Gray points are anatomical context without simulated activity. Silencing a neuron zeroes both its states, and controls replay the same input from zero for an interpretable intervention. The body uses 39 attributed NeuroMechFly meshes and verified articulated kinematics [12]. It is a female body paired with a male brain graph. Head and wing offsets are designed summaries of state, not trained motor outputs, physics, reward learning or animal behavior.

8 Limits and next experiments

The selected central subset changes connectivity and removes biological inputs. Source signs, artificial pooling and rate dynamics simplify biology. The shared vocabulary, short training chunks, modest exposure and two seeds limit language and statistical conclusions. A CPU-friendly implementation may have very different cost on sparse accelerators. No energy efficiency or full-connectome training result is claimed.

Final test scoring freezes checkpoint hashes only after every registered main run completes, scores all test articles and bootstraps paired article loss differences. These intervals are conditional on trained checkpoints; two seeds do not characterize broad training variability. Follow-up controls preserve directed degrees and source-sign constraints through actual edge rewiring, and disable recurrence or slow state under the same schedule before attributing a benefit to anatomy.

Further work should separate improved lexical learning from a graph effect; evaluate delayed recall and context interference; test adaptation retention and reset boundaries; and extend to a provenance-audited small-data corpus such as BabyLM. Whole-graph language training requires a separate sparse-kernel and memory study. Learned motor behavior requires an embodied task, observations, actions and an objective. Those studies may be valuable, but none follows automatically from language tokens coloring a rendered brain.

References

- [1] Guillaume Bellec et al. A solution to the learning dilemma for recurrent networks of spiking neurons. *Nature Communications*, 2020. <https://doi.org/10.1038/s41467-020-17236-y>.
- [2] Kyunghyun Cho et al. Learning phrase representations using RNN encoder-decoder for statistical machine translation. <https://arxiv.org/abs/1406.1078>, 2014.
- [3] Google Research. A connectomics milestone: mapping the complete male fruit-fly brain. <https://research.google/blog/a-connectomics-milestone-mapping-the-complete-male-fruit-fly-brain/>, 2026.
- [4] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. <https://arxiv.org/abs/2312.00752>, 2023.
- [5] Janelia Research Campus. MaleCNS connectome release and downloads. <https://male-cns.janelia.org/download/>, 2026.
- [6] Janne K. Lappalainen et al. Connectome-constrained networks predict neural activity across the fly visual system. *Nature*, 2024. <https://doi.org/10.1038/s41586-024-07939-3>.
- [7] Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. <https://arxiv.org/abs/1609.07843>, 2016.
- [8] Jacob Morra, Andrew Flynn, Andreas Amann, and Mark Daley. Multifunctionality in a connectome-based reservoir computer. <https://arxiv.org/abs/2306.01885>, 2023.
- [9] Bo Peng et al. RWKV: Reinventing RNNs for the transformer era. <https://arxiv.org/abs/2305.13048>, 2023.
- [10] Philip K. Shiu et al. A Drosophila computational brain model reveals sensorimotor processing. *Nature*, 634:210–219, 2024. <https://doi.org/10.1038/s41586-024-07763-9>.
- [11] Ashish Vaswani et al. Attention is all you need. <https://arxiv.org/abs/1706.03762>, 2017.
- [12] S. Wang-Chen et al. NeuroMechFly v2: simulating embodied sensorimotor control in adult Drosophila. *Nature Methods*, 21:2353–2362, 2024. <https://doi.org/10.1038/s41592-024-02497-y>.
- [13] Rui-Jie Zhu et al. SpikeGPT: Generative pre-trained language model with spiking neural networks. <https://arxiv.org/abs/2302.13939>, 2023.